Statement of Research Interests

Overview

I propose to build a research program in **manufacturing and quality science for neuropsychiatric and neurodegenerative cell-based therapeutics and disease models** — developing the biomaterials, biological protocols, and computational infrastructure needed to make iPSC-derived neural and glial cell products reproducible, characterizable, and scalable, from the bench to a manufacturable process.

This is not two adjacent interests bolted together. It is a single problem approached from both ends. Cell and gene therapy for neuropsychiatric and neurodegenerative disease is bottlenecked less by cell biology than by *process*: the field can differentiate iPSCs into disease-relevant neural and glial populations, but struggles to do so reproducibly, at scale, under the kind of quality control a clinical product or a rigorous disease model demands. I have spent the last several years working both halves of that bottleneck — engineering the biomaterials and computational tools that characterize neural cell products, and running the large-scale, quality-controlled operations that make reproducibility at scale a solved problem in other industries. My research program exists to bring those two things together.

Foundational Work

My doctoral research (UTEP, Materials Science & Engineering, 2016–2020, advised by Dr. Binata Joddar) developed an electrically conductive cardiovascular tissue composite — a HiPco single-walled carbon nanotube (SWNT)–alginate hydrogel — as a delivery vehicle for cardiac cell therapy, in collaboration with the Marti Lab at Rice University. I characterized SWNT colloidal stability via zeta-potential measurement across eleven surfactant systems and measured bulk electrical conductivity of the resulting composites (up to ~2.1 mS/cm, crosslinked, Pluronic F108 formulation) — establishing the structure-property relationships needed to engineer a scaffold that is both biocompatible and functionally conductive. This work, along with contributions to engineered bone scaffolds, bioprinted skin constructs, and degradation modeling under physiologic conditions, produced six peer-reviewed publications and trained me in the rigorous, standards-compliant (ASTM/ISO 9001) characterization that translational biomaterials require.

My postdoctoral appointment (Texas Tech University Health Sciences Center El Paso, Neuropsychiatry, 2022) shifted this materials-engineering foundation toward neural cell products directly. I directed non-integrating Sendai-virus reprogramming (CytoTune-iPS 2.0) of patient PBMCs into cGMP-compliant iPSCs for differentiation into neuronal and glial lineages, and led primary tissue processing for the Southwest Brain Bank — papain/DNase dissociation and Neurobasal/B-27/bFGF culture of fresh postmortem brain tissue for cryopreserved biobank quality assessment. In parallel, I led data structuring and biostatistical analysis for transcriptomic studies of neuropsychiatric disease, contributing to a peer-reviewed study of nicotine-exposure gene alterations in human and rodent brain (Vargas-Medrano et al., *Frontiers in Psychiatry*, 2023). That publication, and the postmortem-brain gene-expression dataset behind it, has continued to generate independent analysis on my part: I built a computational structural-pharmacology study of the SERT S438T variant's effect on escitalopram binding (multi-seed AutoDock Vina docking against PDB 5I6Z), an interactive dashboard visualizing oligodendrocyte/myelin gene-expression differences in nicotine-exposed brain tissue, and a statistical analysis of Luminex neuro-inflammatory biomarkers (GFAP, UCHL1, CNTF, and others) across a major depressive disorder and suicidal-behavior cohort (n=70). Each of these was conducted independently,

post-appointment, as a continuation of scientific questions the postdoctoral data raised that I did not want to leave unanswered.

Since 2024, I have managed a 60+ person operations team at Amazon, where I have driven an 82% throughput improvement and maintained 99.7% quality compliance in a highly regulated, high-volume environment — building Python-based forecasting systems, real-time KPI dashboards, and standardized process controls along the way. I do not present this as a detour from research. It is where I learned, at industrial scale, the discipline this field's central bottleneck actually requires: how to define a quality attribute, build the instrumentation to track it, and standardize a process so that its output is the same on day 400 as it was on day one. I have already begun applying this directly to biomanufacturing: I independently built LabOps.AI, a prototype platform that extracts Critical Quality Attributes and Critical Process Parameters from cell/gene-therapy literature (referencing USP <85>/<1116>, ISO 14644-1, and ICH Q6B) into a structured, queryable schema, and I have drafted a review manuscript on the validation gap in AI-guided cell/gene-therapy biomanufacturing and bioprinting. I have also extended my dissertation's bioprinting work into an image-to-scaffold pipeline that converts DICOM imaging data into validated, TPMS-infilled bioprint-ready STL geometry.

Proposed Research Program

Aim 1 — Manufacturing-grade characterization of iPSC-derived neural and glial cell products. I propose to develop standardized protocols and companion analytical tooling for characterizing iPSC-derived neuronal and glial populations against explicit, quantitative critical quality attributes — building directly on my cGMP reprogramming and primary neural culture experience, and extending the CQA/CPP extraction architecture I prototyped in LabOps.AI into a validated framework grounded in real differentiation data rather than literature alone. The goal is a reproducibility standard for neural cell products analogous to what already exists for more mature cell-therapy modalities.

Aim 2 — Computational and biomarker-driven characterization of neuropsychiatric drug targets and glial dysfunction. Building on my SERT/escitalopram docking work, the glial gene-expression dashboard, and the Luminex biomarker analysis, I propose to integrate computational structural pharmacology with transcriptomic and proteomic biomarker profiling to characterize how specific glial and neuronal populations respond to psychiatric therapeutics — generating the functional read-outs needed to validate whether a manufactured neural cell product actually models the disease biology it is meant to represent.

Aim 3 — Scalable, quality-controlled manufacturing of engineered neural tissue constructs. Extending my doctoral work on conductive biomaterial scaffolds and my bioprint-scaffold pipeline, I propose to apply industrial-grade process-control methodology — the kind of standardized, instrumented, quality-tracked process I have run at Amazon — to the manufacture of engineered neural tissue constructs, closing the loop between materials design, bioprinting, and reproducible, scalable output.

Fit and Long-Term Vision

Georgia Tech's Coulter Department, through the NSF Engineering Research Center for Cell Manufacturing Technologies (CMaT) and the Marcus Center for Therapeutic Cell Characterization and Manufacturing (MC3M), is the natural home for Aims 1 and 3; the department's growing neuroscience and neuroengineering infrastructure — the GT/Emory Neural Engineering Center and the Institute for

Neuroscience, Neurotechnology, and Society — offers real collaborative depth for Aim 2. My long-term goal is a research program that treats "manufacturing" and "neuroscience" as one discipline where neuropsychiatric and neurodegenerative cell therapies are concerned, training students who are as comfortable with a quality system as they are with a differentiation protocol — because for this class of therapeutics, I do not believe the field will get one without the other.

I plan to pursue this program through NIH (NIBIB for biomanufacturing technology development; NIMH/NINDS for the neuropsychiatric and neurodegenerative disease-modeling components), NSF (Future Manufacturing and Engineering Research Center-adjacent mechanisms), ARPA-H, and Georgia Research Alliance seed funding, alongside industry partnerships with cell-therapy CDMOs for whom this reproducibility problem is a direct commercial concern.